

THOMAS TO

Oakland, CA | 510-387-5408 | thomas.to.bcheme@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer and Full Stack Software Engineer with 5+ years building production ML systems, Retrieval-Augmented Generation (RAG) pipelines, and large language model (LLM) integrations. Designs React and Next.js frontends, Python and FastAPI backend services, and RESTful APIs exposing ML capabilities to end users. Deploys containerized MLOps pipelines on GCP and AWS with CI/CD automation, prompt engineering, and fine-tuning for domain-specific LLM accuracy. Collaborates cross-functionally with scientific and executive stakeholders, mentors junior engineers, and delivers iterative improvements tied to business outcomes.

TECHNICAL SKILLS

- **AI and ML:** Python, Large Language Models (OpenAI GPT, Anthropic Claude, Google Gemini), RAG, PyTorch, TensorFlow, scikit-learn, Fine-tuning (PEFT/LoRA), LangChain, LangGraph, Multi-Agent Orchestration, Prompt Engineering, Vector Databases (Pinecone), Hugging Face, MCP, n8n
- **Languages:** Python (Pandas, NumPy, PySpark), TypeScript, JavaScript, SQL, R
- **Web and APIs:** React, Next.js, FastAPI, Node.js, RESTful APIs, Vue.js, Tailwind CSS, Streamlit
- **Cloud and DevOps:** GCP, AWS (DynamoDB, S3), Docker, Kubernetes, CI/CD, MLOps, GitHub Actions, Vertex AI, Vercel, Model Monitoring
- **Data Management:** Snowflake, PostgreSQL, MongoDB, dbt, Fivetran, ETL/ELT Pipelines, Data Warehousing, Tableau, SAP S/4HANA

PROFESSIONAL EXPERIENCE

Master Data Specialist | Canvanta Life Sciences | Emeryville, CA | Oct 2023 - Present

- Reduced stakeholder decision-making from hours to minutes, by architecting a revenue optimization system integrating a predictive ML model with a RAG AI agent on Snowflake, eliminating manual review cycles.
- Achieved 95%+ accuracy on legacy data digitization, by fine-tuning large language models on 5+ years of handwritten documents through systematic prompt engineering and evaluation.
- Reduced daily calculation time by 87% (-40 min) and forecasted production within 3 units of actual, by deploying a fullstack platform on GCP via CI/CD with Docker, converting serial to concurrent workflows.

AI Engineer (Open Source and R&D) | Oakland, CA | Oct 2023 - Present

- Deployed an end-to-end MLOps pipeline spanning frontend to model tracking, by building a React and Next.js frontend on AWS, Python ML models (scikit-learn, PyTorch, TensorFlow) on Hugging Face via FastAPI and Docker, and model tracking via GCP Vertex AI.
- Improved domain-specific LLM accuracy for production workflows, by building multi-agent orchestration systems using LangChain and LangGraph with structured prompt engineering for RAG pipelines.

Software Engineer | Genentech | South San Francisco, CA | Jun 2022 - Jan 2023

- Streamlined cross-functional workflows for medical writing and scientific stakeholder teams, by engineering full-stack applications with a Python REST API backend and React, Vue.js, and Node.js frontend.
- Improved document retrieval speed by 60%, by designing knowledge management platforms with MongoDB and PostgreSQL through query optimization and structured data modeling.

Research Engineer | UC Davis (Nandi, McDonald, Wan, Siegel Labs) | Davis, CA | Sep 2019 - Jun 2021

- Reduced manufacturing costs by \$63.2 Million through performance-critical improvements, by designing computational models in Python (NumPy, Pandas) with numerical algorithms and parallel execution.
- Enabled high-throughput quantitative analysis from large experimental datasets, by building Python computer vision pipelines (OpenCV, ImageJ) for automated microscopy image analysis to quantify organoid growth.

EDUCATION

Bachelor of Science, Biochemical Engineering | University of California, Davis | Davis, CA | Sep 2019 - Jun 2022

- AWS Cloud Technical Essentials
- Advanced Machine Learning with Agentic AI (ongoing)